

Numerical Normality Audit of the Exponentially Weighted Endpoint Coordinate

Finite-lookback scale calibration, Q-Q geometry, and tail behavior

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Reproducible computational supplement

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Abstract

This report asks a finite-sample question not settled by asymptotic normality alone. For the stationary log-odds endpoint coordinate $L_t = \log(U_t/D_t)$, does standardization by the asymptotic scale or by the derived law-specific finite-lookback correction produce an approximately standard normal marginal distribution at practically relevant lookbacks?

The audit evaluates 216 million coordinate values: 3 million retained time points for each of 72 law-lookback combinations. The laws are Gaussian, Laplace, and standardized Student t_ν with $\nu = 4, 5, 6, 8$; the lookbacks are 2, 3, 5, 8, 13, 21, 35, 55, 89, 144, 233, 377. The analysis reports variance, skewness, excess kurtosis, Kolmogorov distance, truncated quantile-grid mean absolute error, normal Q-Q regressions, central coverage, and tail frequencies. An empirical variance “oracle” is included only to distinguish scale error from residual shape error.

The result is not exact finite- n Gaussianity. The law-specific correction usually removes most of the scale error, and the remaining shape approaches a Gaussian form with increasing lookback. Gaussian and Laplace cases are already close in Kolmogorov distance at short horizons, although their positive excess kurtosis remains visible. Student t_4 is the important short-horizon exception: its first shift vanishes ($K_4 = 0$), so the first-order corrected scale coincides with the asymptotic scale and substantially over-scales $n = 2$ and $n = 3$. Across all laws the Q-Q grids become visually close to linear at moderate lookbacks, but variance normalization should not be described as an exact Gaussianizing transform.

1 Question and answer

The analytical manuscript proves an asymptotic Gaussian limit and a law-dependent finite-lookback variance expansion under its stated assumptions. Neither statement says that the transformed statistic is exactly normal at any fixed lookback. This audit therefore separates three questions:

1. Does the proposed scale make the variance close to one?
2. After removing scale error, is the remaining distributional shape close to standard normal?
3. Is the resulting distribution stable enough across lookbacks to support percentile comparisons?

The numerical answer is:

- **Scale:** the law-specific correction is much better than the first-order asymptotic scale at short and moderate lookbacks for Gaussian and Laplace innovations. The Student first-shift correction can be good, poor, or directionally wrong at $n = 2$, depending on ν .
- **Shape:** finite- n distributions retain excess kurtosis. The deviation is modest for Gaussian and Laplace inputs and decreases with n . The oracle diagnostic confirms that standardizing the variance does not eliminate all non-Gaussian shape.
- **Use:** the coordinate is an increasingly accurate Gaussian pivot, not an exact small-sample pivot. For practical small- n thresholds, law-specific empirical percentiles remain safer than universal normal cutoffs.

2 Statistic and scales

For gain $\alpha = 1/n$, let

$$U_t = (1 - \alpha)U_{t-1} + \alpha \max(\varepsilon_t, 0), \quad (1)$$

$$D_t = (1 - \alpha)D_{t-1} + \alpha \max(-\varepsilon_t, 0), \quad (2)$$

and define

$$L_t^{(n)} = \log \frac{U_t}{D_t}. \quad (3)$$

For a centered innovation law with standard deviation σ and $m_1 = \mathbb{E}|\varepsilon|$, the asymptotic scale is

$$C_F = \frac{\sqrt{2}\sigma}{m_1}. \quad (4)$$

Three normalizations are compared:

$$Z_n^{\text{asym}} = \frac{\sqrt{n}}{C_F} L_t^{(n)}, \quad (5)$$

$$Z_n^{\text{corr}} = \frac{\sqrt{g_F(n)}}{C_F} L_t^{(n)}, \quad (6)$$

$$Z_n^{\text{oracle}} = \frac{L_t^{(n)} - \bar{L}_n}{s(L_n)}. \quad (7)$$

The oracle is not a proposed normalization. It uses the same simulation sample to center and scale the statistic and therefore answers only whether the shape, after scale removal, resembles a normal distribution.

For Gaussian and Laplace laws, the correction uses

$$g_F(n) = n - K_F + \frac{A_F}{n} + \frac{B_F}{n^2}. \quad (8)$$

For Student laws it uses only the first-shift correction developed in the companion conventional technical note, $g_\nu(n) = n - K_\nu$. This correction remains outside Lean, and that distinction is material at small n .

Table 1: Scale constants and correction formulas used by the audit.

Law	C_F	K_F	Correction used
Gaussian	1.772454	1.593657	$n - K_F + A_F/n + B_F/n^2$
Laplace	2.000000	1.333333	$n - K_F + A_F/n + B_F/n^2$
Student-t4	2.000000	-0.000000	$n - K_F$ (analytic note)
Student-t5	1.923825	0.959710	$n - K_F$ (analytic note)
Student-t6	1.885618	1.222222	$n - K_F$ (analytic note)
Student-t8	1.847521	1.400000	$n - K_F$ (analytic note)

3 Simulation design

3.1 Path construction

For each innovation law, one reproducible path is generated with seed 20260923. The same innovation path is used across lookbacks within a law, making comparisons paired. The Wilder recursions start from zero and the first $32 \max(n) = 12064$ observations are discarded. At $n = 377$, the remaining weight of the zero initialization is below approximately $\exp(-32)$; it is smaller for every other lookback.

After burn-in, 3,000,000 time points are retained for every law-lookback pair. These observations represent the stationary marginal distribution but are not independent. Standard i.i.d. normality-test p-values would therefore be miscalibrated and, at this sample size, scientifically unhelpful. They are not reported.

3.2 Metrics

The audit reports:

- mean, variance, skewness, and excess kurtosis;
- one-sample Kolmogorov distance from $N(0, 1)$;
- mean absolute error on a dense normal-quantile grid over probabilities 0.005 through 0.995. This is a truncated quantile-grid diagnostic, stored as `wasserstein_grid`, not the full W_1 distance on the real line;
- a linear Q-Q fit over probabilities 0.005 through 0.995: intercept, slope, and R^2 ;
- coverage of the nominal 95% and 99% normal intervals;
- $\mathbb{P}(|Z| > 2)$ and $\mathbb{P}(|Z| > 3)$;
- central-quantile RMSE relative to the $n = 377$ distribution.

The path is partitioned into 192 contiguous blocks. Block-to-block variation provides correlation-aware Monte Carlo standard errors for moments, coverage, and tail frequencies. The longest-lookback block contains more than 41 lookbacks. These errors measure simulation stability; they are not a claim that the blocks are exactly independent.

Complete aggregate outputs, quantile grids, block summaries, and representative draws are provided in the accompanying reproducibility package; see Appendix B.

4 Validation of the computation

The computational validation verifies all of the following:

- all 216 law-lookback-normalizer rows are present with unique keys;
- every required numerical field is finite;
- the reported variances satisfy the scale identity to a maximum absolute error of 1.34×10^{-15} ;
- no comparison between the first and second halves of the block sequence exceeds three pooled block standard errors. The maximum absolute split diagnostic is 2.82.

These checks assess completeness, algebraic consistency, and simulation stability; they are not a formal proof of the model or unadjusted multiple hypothesis tests. The machine-readable checks and file inventory are documented in Appendix B and in the repository.

5 Results

5.1 Variance calibration

Figure 1 compares the asymptotic and finite- n scales. For Gaussian and Laplace inputs the corrected variance is materially closer to one over most of the grid. At $n = 2$, the Gaussian fourth-order approximation still leaves variance 1.0847, while the Laplace value is 1.0084. For Laplace innovations, a small residual over-dispersion of approximately 1–2% remains across much of the moderate-lookback grid. This is a useful warning: a fourth-order large- n expansion can be accurate at a very small horizon without being an exact finite- n calibration.

For Student inputs, the first shift is visibly asymptotic. At $n = 2$, the corrected variances are 2.2672, 1.2589, 0.9670, and 0.7719 for $\nu = 4, 5, 6, 8$, respectively. The fact that t_6 happens to be close to one does not make the formula uniformly small-sample accurate across ν . For $\nu = 4$, $K_4 = 0$ and hence $g_4(n) = n$: the corrected and asymptotic normalizations are identical throughout the grid. Its small- n over-dispersion therefore records a limitation of the first-shift approximation, not the effect of a distinct finite- n rescaling.

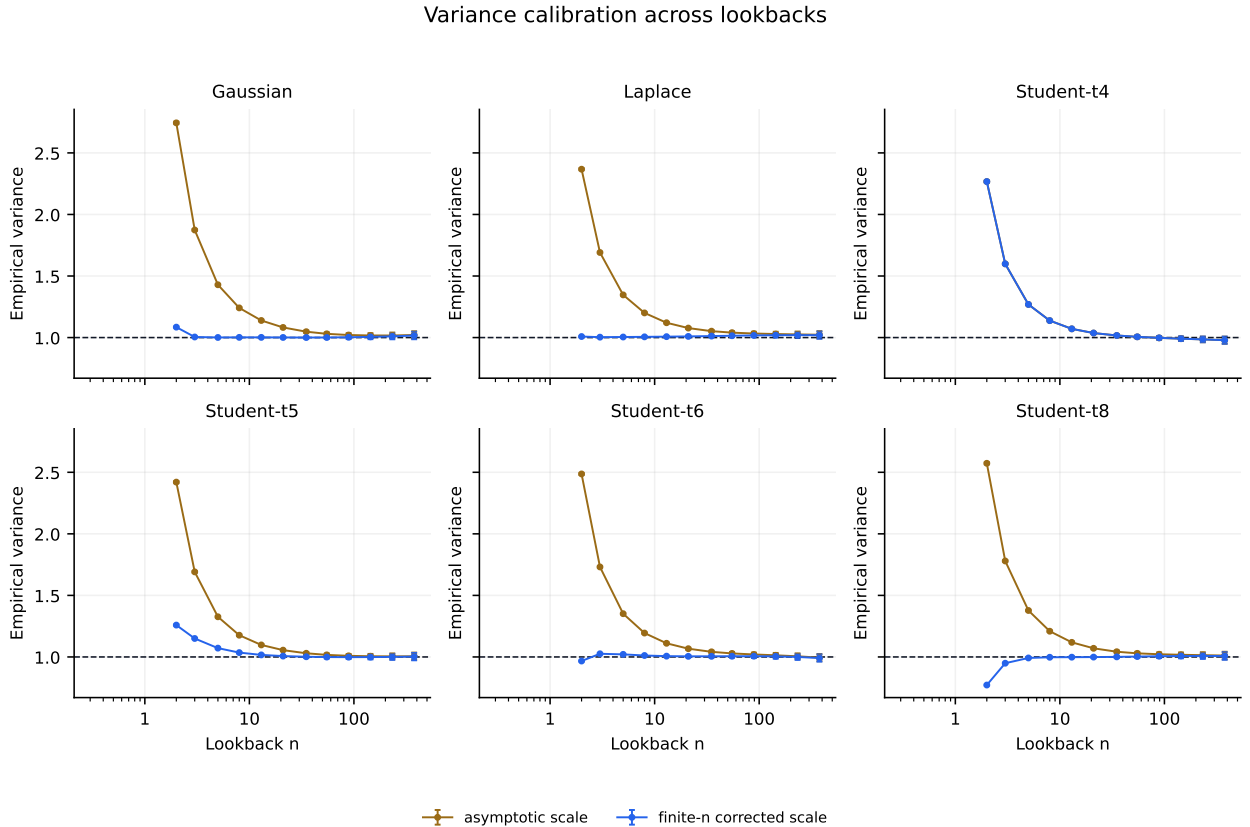


Figure 1: Empirical variance under the first-order asymptotic scale and the law-specific finite-lookback scale. Error bars are 95% block Monte Carlo intervals. The oracle is omitted because its sample variance equals one by construction.

5.2 Scale error versus shape error

At $n = 2$, the asymptotic scale produces Kolmogorov distances near 0.09–0.11 for every law. The finite- n correction reduces that distance to 0.0056 for Gaussian and 0.0077 for Laplace. Student results depend strongly on ν : 0.0879 for t_4 , 0.0184 for t_5 , 0.0143 for t_6 , and 0.0417 for t_8 .

After empirical variance normalization, all six $n = 2$ Kolmogorov distances fall in the narrow interval 0.0087–0.0113. This is the clearest diagnostic separation in the study. Most of the dramatic Student t_4 error is scale error, not an exceptionally distorted central shape. Nevertheless, all six laws have positive excess kurtosis between 0.54 and 0.67 at $n = 2$, so the oracle distributions are not exactly Gaussian.

Table 2: At $n = 2$, comparison of scale error with residual shape. The oracle column uses sample centering and variance and is diagnostic only.

Law	Var corr.	KS asym.	KS corr.	KS oracle	Kurtosis
Gaussian	1.0847	0.1087	0.0056	0.0113	0.6665
Laplace	1.0084	0.0945	0.0077	0.0087	0.5355
Student- t_4	2.2672	0.0879	0.0879	0.0101	0.6256
Student- t_5	1.2589	0.0953	0.0184	0.0104	0.6172
Student- t_6	0.9670	0.0986	0.0143	0.0103	0.6098
Student- t_8	0.7719	0.1021	0.0417	0.0107	0.6367

Scale correction versus residual distributional shape

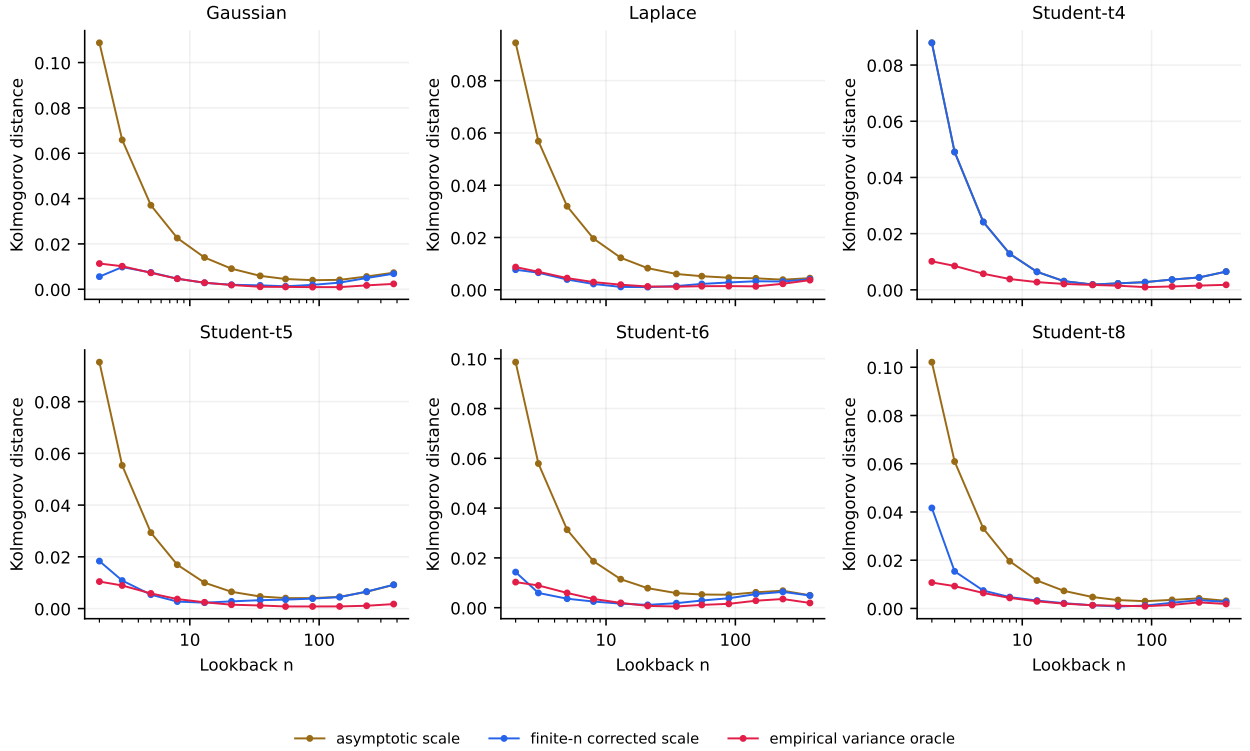


Figure 2: Kolmogorov distance for the asymptotic, finite- n corrected, and oracle scales. The gap between corrected and oracle curves measures scale error; the oracle curve retains pure finite- n shape error plus Monte Carlo noise.

5.3 Convergence with lookback

By $n = 13$, the corrected Kolmogorov distances are 0.0028 (Gaussian), 0.0011 (Laplace), 0.0064 (Student t_4), 0.0023 (Student t_5), 0.0016 (Student t_6), and 0.0033 (Student t_8). By $n = 55$, every corrected distance is below 0.0035. At the largest lookbacks the descriptive distance can rise slightly. This is consistent with the decreasing effective number of nearly independent states and with Monte Carlo variability in both variance and the empirical

CDF. The oracle and block diagnostics do not isolate a single cause, but the observed rise is not evidence of a reversal of the asymptotic result.

The Q-Q regressions are almost perfectly linear in the center even at small lookbacks. At $n = 2$, corrected Q-Q slopes range from 0.868 for Student t_8 to 1.488 for Student t_4 , while every reported central-grid R^2 exceeds 0.998. Thus a visually straight Q-Q line can coexist with a materially wrong scale and non-Gaussian tails. Slope, kurtosis, coverage, and tail frequency must be read together.

Finite-n Gaussian approximation after law-specific scaling

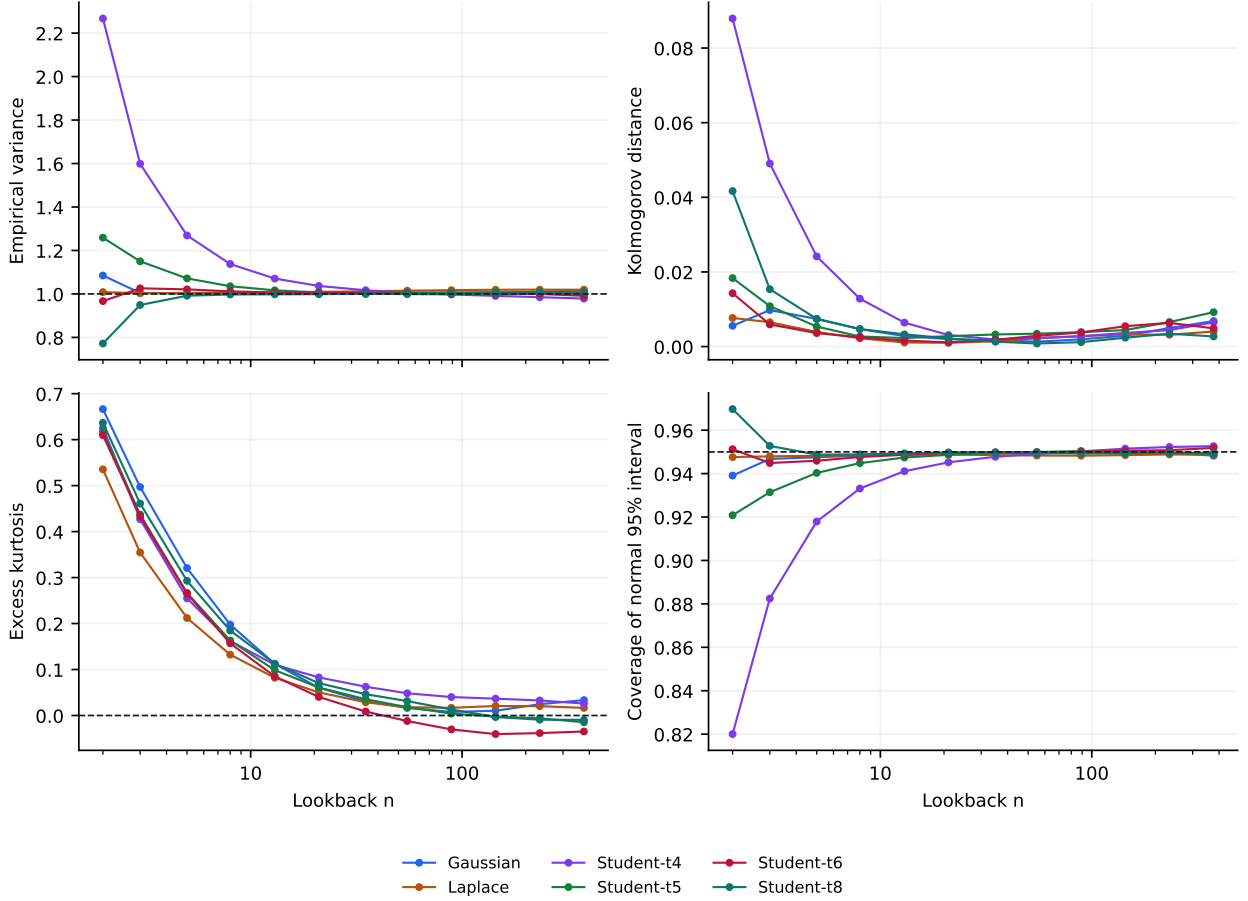


Figure 3: Variance, Kolmogorov distance, excess kurtosis, and nominal 95% coverage after the finite-lookback correction. The dashed lines mark standard normal reference values where applicable.

5.4 Tail behavior

The standard normal benchmark is $\mathbb{P}(|Z| > 3) \approx 0.00270$. Gaussian input at $n = 2$ produces 0.00819 after correction, and Laplace input produces 0.00589. Student t_4 reaches 0.04879 because its small- n scale is severely over-dispersed. At moderate lookbacks the values move toward the normal benchmark, with residual differences that agree with the excess-kurtosis pattern.

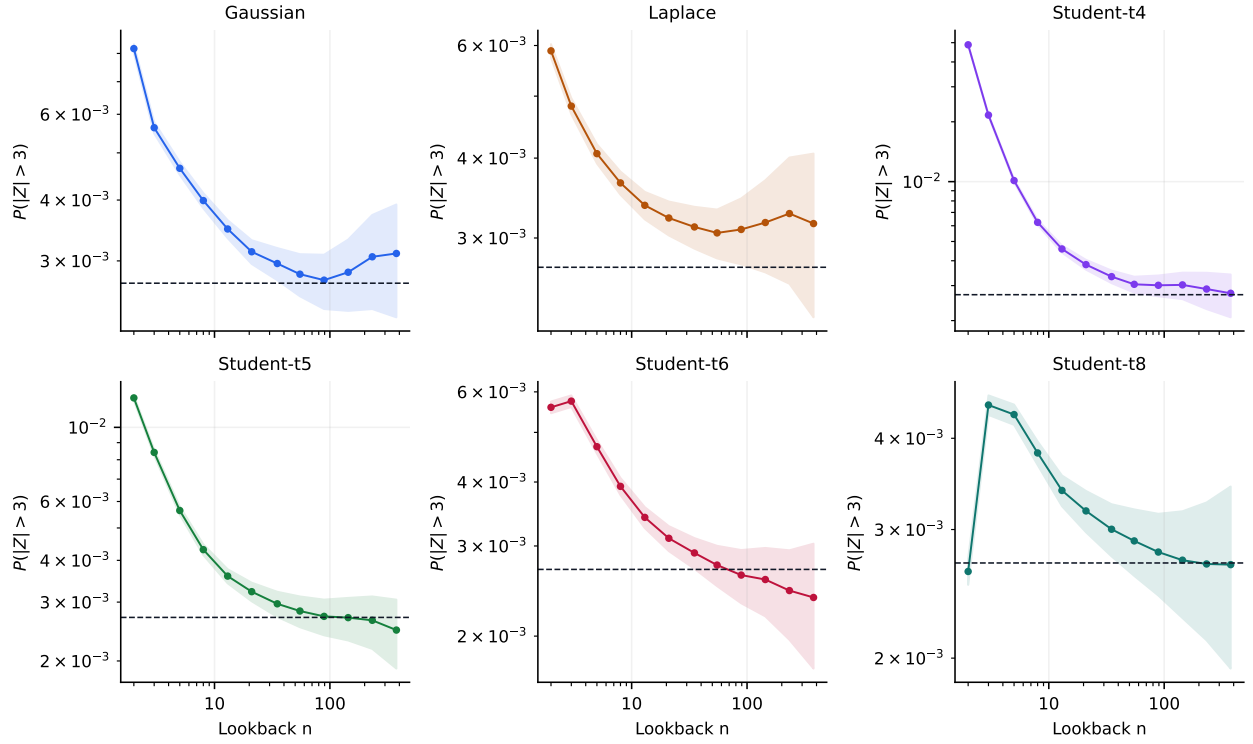
Three-sigma tail calibration after finite- n scaling

Figure 4: Three-sigma tail frequencies on a logarithmic scale. Shading shows 95% block Monte Carlo intervals; the dashed line is the standard normal tail probability.

5.5 Percentile stability

The percentile heatmaps show empirical minus normal quantiles after the finite-lookback correction. Warm and cool bands at $n = 2$ reflect the joint effect of scale error and finite- n tail shape. Errors contract rapidly for Gaussian and Laplace inputs. Student t_4 needs a longer horizon because the first-shift correction leaves substantial variance error at the smallest lookbacks.

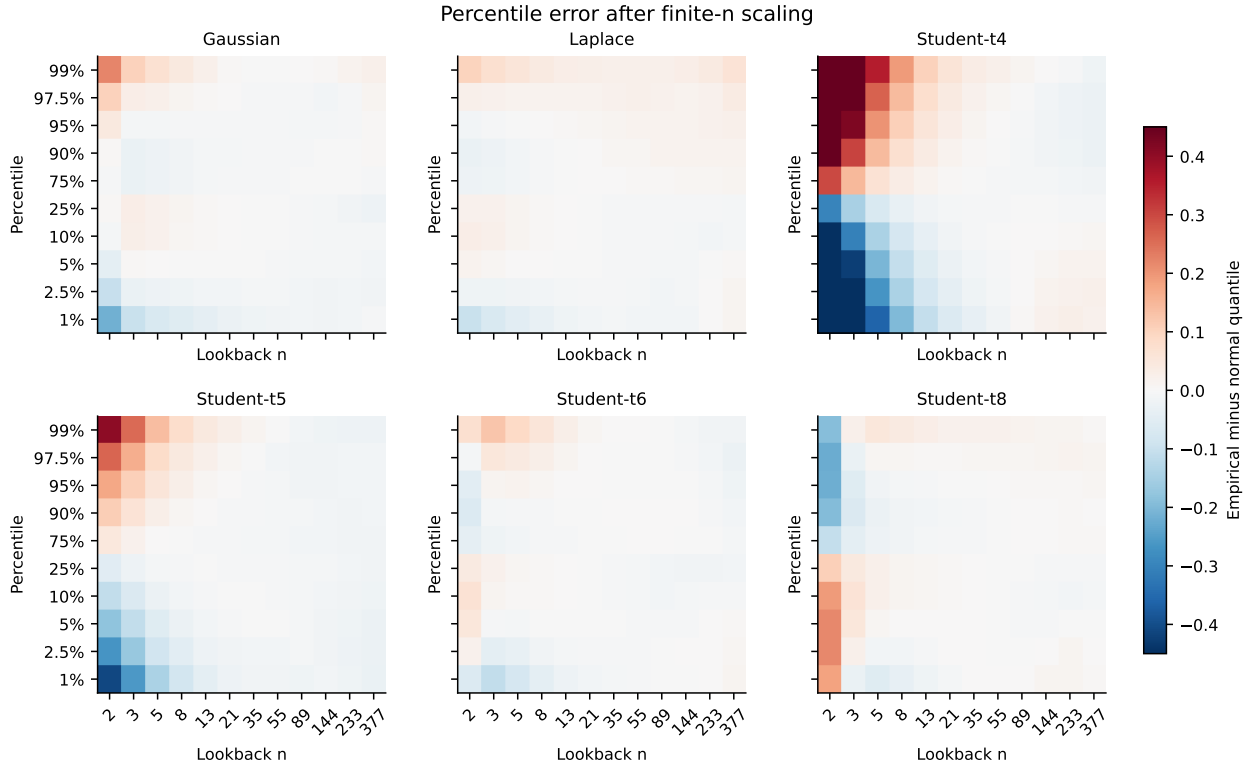


Figure 5: Difference between empirical corrected quantiles and standard normal quantiles. Red is above and blue below the normal reference.

Table 3: Selected results for the law-specific finite- n scaling. Block MCSE is shown for the variance.

Law	n	$\text{Var}(Z)$	MCSE	KS	Excess kurt.	Coverage 95%	$P(Z > 3)$
Gaussian	2	1.0847	0.0020	0.0056	0.6665	0.9391	0.0082
	5	1.0008	0.0020	0.0074	0.3207	0.9475	0.0046
	13	1.0016	0.0028	0.0028	0.1123	0.9489	0.0035
	55	1.0006	0.0058	0.0013	0.0164	0.9498	0.0028
	377	1.0166	0.0154	0.0068	0.0335	0.9483	0.0031
Laplace	2	1.0084	0.0017	0.0077	0.5355	0.9476	0.0059
	5	1.0045	0.0020	0.0039	0.2121	0.9482	0.0041
	13	1.0074	0.0028	0.0011	0.0821	0.9487	0.0034
	55	1.0149	0.0056	0.0022	0.0180	0.9483	0.0031
	377	1.0194	0.0149	0.0040	0.0165	0.9487	0.0032
Student-t4	2	2.2672	0.0040	0.0879	0.6256	0.8200	0.0488
	5	1.2691	0.0024	0.0242	0.2548	0.9179	0.0101
	13	1.0708	0.0030	0.0064	0.1104	0.9411	0.0046
	55	1.0061	0.0058	0.0023	0.0484	0.9491	0.0030
	377	0.9787	0.0147	0.0065	0.0260	0.9526	0.0027
Student-t5	2	1.2589	0.0023	0.0184	0.6172	0.9208	0.0122
	5	1.0717	0.0022	0.0054	0.2667	0.9403	0.0056
	13	1.0166	0.0030	0.0023	0.0986	0.9474	0.0036
	55	0.9991	0.0058	0.0035	0.0182	0.9499	0.0028
	377	1.0028	0.0152	0.0092	-0.0147	0.9491	0.0025
Student-t6	2	0.9670	0.0017	0.0143	0.6098	0.9513	0.0056
	5	1.0210	0.0020	0.0036	0.2651	0.9459	0.0047
	13	1.0066	0.0030	0.0016	0.0851	0.9487	0.0034
	55	1.0059	0.0059	0.0029	-0.0120	0.9498	0.0028
	377	0.9913	0.0148	0.0049	-0.0348	0.9518	0.0024

Law	n	$\text{Var}(Z)$	MCSE	KS	Excess kurt.	Coverage 95%	$P(Z > 3)$
Student-t8	2	0.7719	0.0014	0.0417	0.6367	0.9697	0.0026
	5	0.9918	0.0020	0.0074	0.2929	0.9487	0.0043
	13	0.9982	0.0030	0.0033	0.1126	0.9493	0.0034
	55	1.0032	0.0060	0.0008	0.0314	0.9494	0.0029
	377	1.0077	0.0156	0.0027	-0.0094	0.9495	0.0027

6 Limitations

- The study evaluates stationary marginal distributions under i.i.d. innovations. It does not test serially dependent, heteroskedastic, or empirical market returns.
- The full retained path is serially dependent. Block errors and split-half checks account for this approximately, but the reported Kolmogorov distances remain descriptive Monte Carlo estimates.
- The correction parameters are not fitted to the simulated data. The oracle is fitted by construction and must not be compared as a deployable method.
- Only symmetric laws are included. Asymmetric increments require centering corrections and are outside the present variance expansion.
- The Student finite- n correction is only first order and is explicitly not promoted to the status of the Gaussian/Laplace fourth-order result.

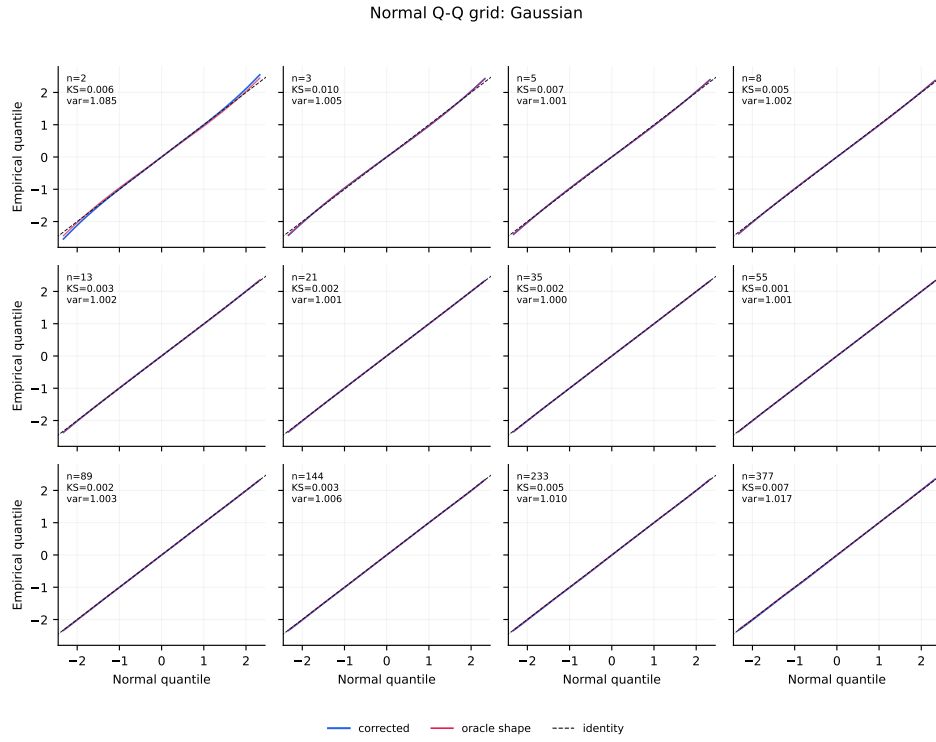
7 Conclusion

The simulations support the asymptotic Gaussian interpretation of the standardized endpoint coordinate while delineating its finite-lookback limits. For Gaussian and Laplace innovations, the law-specific corrections materially improve variance calibration, including at short lookbacks in this experiment. They do not, however, produce exact finite- n Gaussianity: residual excess kurtosis and tail discrepancies remain visible after scale error is removed.

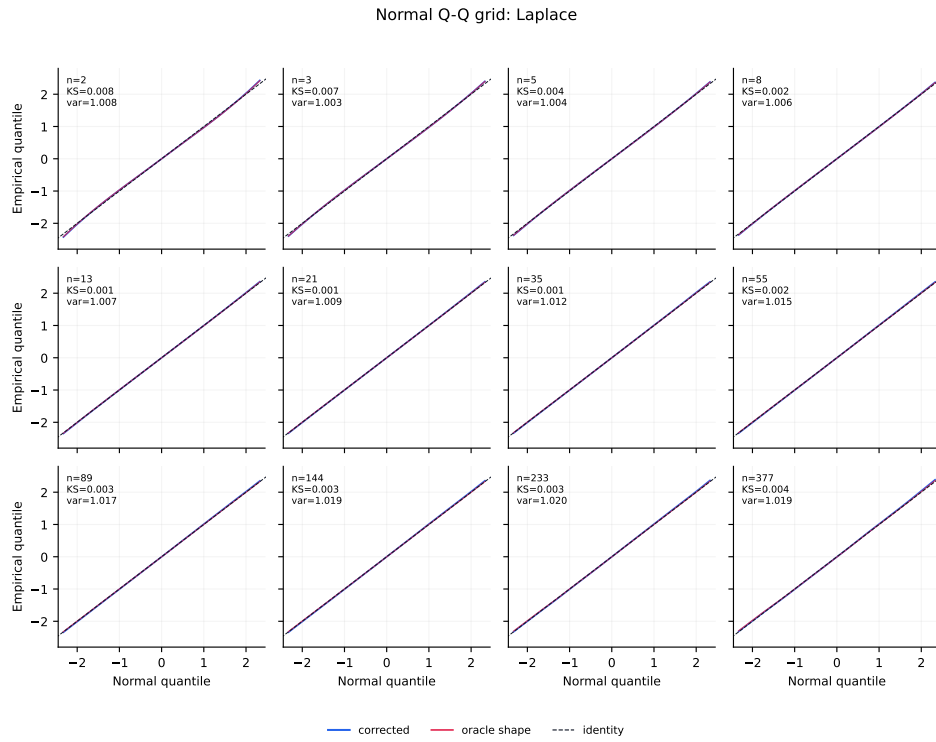
The Student first-shift formula remains a first-order analytic correction outside Lean. In particular, its short-horizon behavior for t_4 reflects the order of approximation rather than a contradiction of the large- n theory. At very small lookbacks, law-specific or empirical percentiles may therefore be more reliable than universal normal cutoffs. The normal approximation becomes increasingly accurate as the lookback grows, at a rate that depends on the innovation law and the tail accuracy required.

A Full Q-Q grids

Each panel compares 99 empirical quantiles with standard normal quantiles. The blue curve uses the proposed finite- n scale. The red curve uses empirical centering and variance and therefore isolates residual shape. The dashed line is exact normal agreement. Panel annotations give corrected variance and Kolmogorov distance.

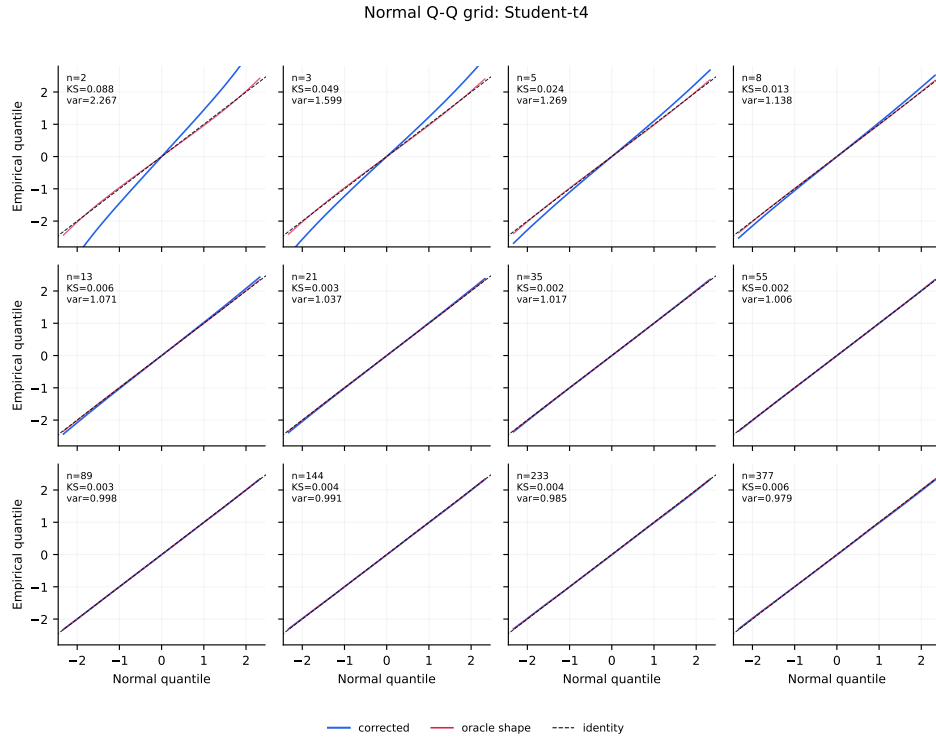
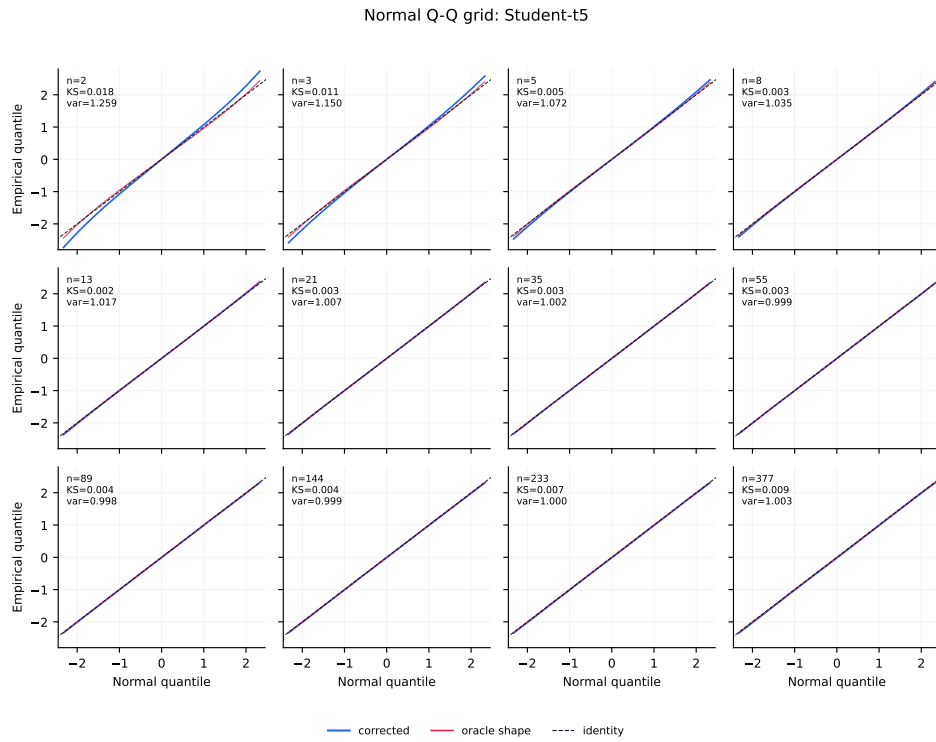


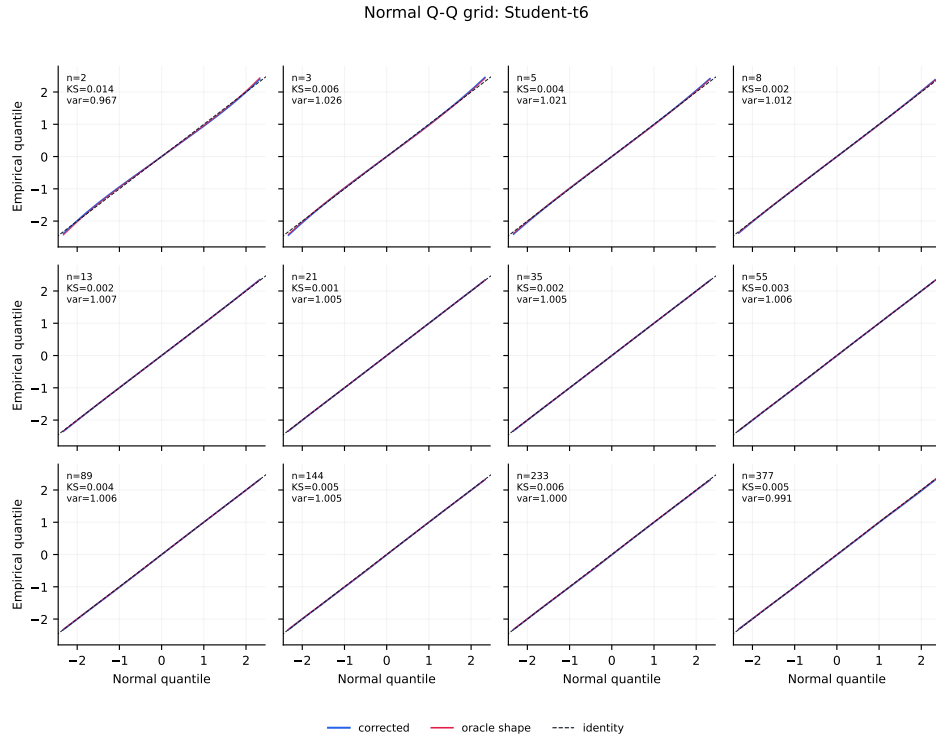
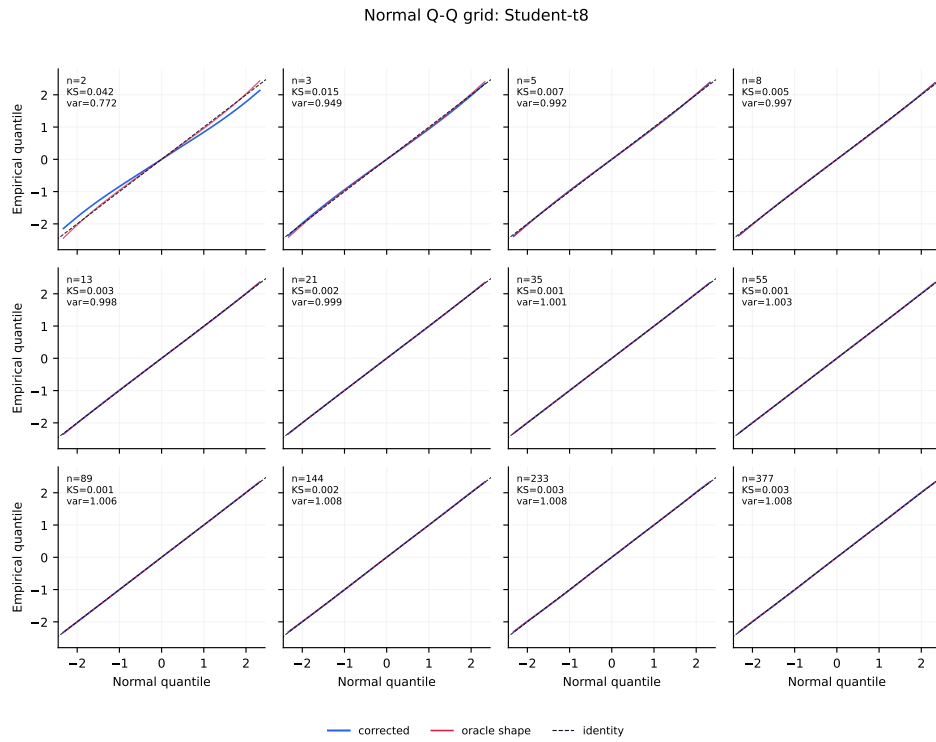
(a) Gaussian innovations.



(b) Laplace innovations.

Figure 6: Normal Q-Q grids for Gaussian and Laplace innovations.

(a) Student t_4 innovations.(b) Student t_5 innovations.Figure 7: Normal Q-Q grids for Student t_4 and t_5 innovations.

(a) Student t_6 innovations.(b) Student t_8 innovations.Figure 8: Normal Q-Q grids for Student t_6 and t_8 innovations.

B Reproducibility and provenance

On Windows, run the complete audit from the repository root with:

```
py -m pip install -r supplement\normality_audit\requirements.txt
py supplement\normality_audit\run_normality_audit.py
```

On Linux or macOS, use:

```
python3 -m pip install -r supplement/normality_audit/requirements.txt
python3 supplement/normality_audit/run_normality_audit.py
```

The archived reproducibility package contains summary metrics, quantile and empirical-CDF grids, histogram counts, block summaries, representative draws, the simulation configuration, machine-readable quality checks, tables, figures, and source files. All figures can be regenerated from the archived aggregate outputs without rerunning the simulation:

```
py supplement\normality_audit\run_normality_audit.py --plots-only
```

On Linux or macOS, replace `py` with `python3` and use forward slashes in the same command. The author formulated the research question, mathematical framework, and interpretation, and takes responsibility for the claims made in this report. AI tools assisted with computational design, code drafting, debugging, visualization, and editorial preparation. Numerical results were produced by the archived code and preserved outputs; they are not Lean-verified theorems.